1.Straightforward Prompts:

Prompt 1 : "what is prompt engineering"

Simple Definition

Prompt engineering means **carefully wording a question or instruction so an AI understands exactly what you want and gives a better answer**.

Example

Normal Prompt:

Explain blockchain.

Engineered Prompt:

Explain blockchain technology in 100 words with a simple example suitable for beginners.

The second prompt gives a **clearer instruction**, so the AI produces a **more accurate and structured response**.

Prompt 2: "list out the different prompt parameters"

Different **prompt parameters** are elements used to control how an AI generates responses.

Prompt Parameters

1. **Instruction** – The task or command given to the AI.
   Example: "Explain blockchain technology."
2. **Context** – Background information that helps the AI understand the situation.
   Example: "Explain blockchain for computer science students."
3. **Input Data** – Specific data or information provided for processing.
   Example: A paragraph that needs summarization.
4. **Output Format** – Specifies how the response should be structured.
   Example: "Provide the answer in a table."
5. **Constraints** – Limits such as word count, tone, or style.
   Example: "Explain in 100 words."

6. **Role/Persona** – Assigning the AI a role to improve the response.
   Example: "Act as a cybersecurity expert."
7. **Examples (Few-shot prompting)** – Providing sample inputs and outputs to guide the AI.
8. **Temperature** – Controls creativity/randomness of the response.
9. **Max Tokens** – Limits the length of the generated output.
10. **Top-p (Nucleus Sampling)** – Controls diversity of generated text.

2.Tabular Format Prompting

Prompt 1 : "create a table of comparing dynamic and static memory allocation in C programming"

Feature	Static Memory Allocation	Dynamic Memory Allocation
Definition	Memory is allocated at compile time.	Memory is allocated during runtime.
Memory Location	Stored in stack memory.	Stored in heap memory.
Size Flexibility	Size must be fixed before program execution.	Size can be changed during program execution.
Allocation Method	Using normal variable declaration.	Using functions like malloc(), calloc(), realloc(), and free().
Memory Management	Managed automatically by the compiler.	Managed manually by the programmer.
Efficiency	Faster allocation and deallocation.	Slightly slower due to runtime allocation.
Example	int arr[10];	int *arr = (int*)malloc(10 * sizeof(int));

Prompt 2 : Provide a table listing five types of databases, their data model, and one example of each.

Database Type	Data Model	Example
Relational Database	Table-based (rows and columns)	MySQL
Hierarchical Database	Tree structure (parent–child relationship)	IBM Information Management System
Network Database	Graph structure with many-to-many relationships	Integrated Data Store
Object-Oriented Database	Stores data as objects similar to OOP	ObjectDB
NoSQL Database	Non-tabular (key-value, document, etc.)	MongoDB

3.Preceding Question Prompting

Prompt 1:

What is artificial intelligence?

Artificial Intelligence (AI) is the ability of a computer or machine to perform tasks that normally require human intelligence. These tasks include learning from data, recognizing patterns, solving problems, understanding language, and making decisions.

AI systems use techniques from fields like Machine Learning, Natural Language Processing, and Computer Vision to improve their performance over time.

Examples of AI in real life:

- Voice assistants like Siri and Google Assistant
- Recommendation systems used by Netflix and Amazon
- Self-driving technology developed by Tesla

In simple words:
Artificial Intelligence is technology that allows machines to **think, learn, and make decisions similar to humans**.

Where is AI commonly used today?

Artificial Intelligence (AI) is commonly used in many areas today to make tasks faster, smarter, and more efficient.

1. Virtual Assistants

AI powers voice assistants like Siri, Google Assistant, and Amazon Alexa that can answer questions, set reminders, and control smart devices.

2. Entertainment and Recommendations

Platforms like Netflix, YouTube, and Spotify use AI to recommend movies, videos, and music based on user preferences.

3. Healthcare

AI helps doctors analyze medical images, detect diseases early, and assist in research for better treatments.

4. Self-Driving and Smart Vehicles

Companies like Tesla use AI to develop autonomous driving systems that help cars drive safely.

5. Online Shopping

E-commerce companies like Amazon use AI for product recommendations, chatbots, and fraud detection.

6. Security and Cybersecurity

AI helps detect unusual activity, prevent hacking, and improve online security systems.

Now explain how AI is used in healthcare.

Artificial Intelligence (AI) is widely used in **healthcare** to improve diagnosis, treatment, and patient care. 🏥🤖

1. Disease Diagnosis

AI helps doctors analyze medical images such as X-rays, CT scans, and MRIs to detect diseases like Cancer or COVID-19 earlier and more accurately.

2. Medical Research and Drug Discovery

AI speeds up the process of discovering new medicines by analyzing large amounts of biological and medical data.

3. Virtual Health Assistants

AI-powered chatbots and apps provide basic medical advice, remind patients to take medicines, and monitor health conditions.

4. Predicting Health Risks

AI systems analyze patient records to predict diseases such as Heart Disease or Diabetes, helping doctors take preventive action early.

5. Robotic Surgery

Advanced AI-powered surgical systems like da Vinci Surgical System assist surgeons in performing precise and minimally invasive operations.

✅ **In simple terms:**
AI helps healthcare professionals **diagnose diseases faster, develop new treatments, and provide better patient care.**

Prompt 2:

What is renewable energy?

Renewable energy is energy that comes from natural sources that are **continuously replenished** and do not run out when used. These sources are naturally renewed by processes in nature.

Examples of Renewable Energy

1. **Solar energy** – energy from the Sun
2. **Wind energy** – energy produced by moving air
3. **Hydropower** – energy from flowing water
4. **Biomass energy** – energy from organic materials like plants and waste
5. **Geothermal energy** – energy from heat inside the Earth

Key Features

- Environmentally friendly 🌱
- Produces little or no pollution
- Sustainable and available for a long time

Example: Solar panels convert sunlight into electricity for homes and buildings.

If you want, I can also explain **the advantages and disadvantages of renewable energy** or give **examples used in India**.

Why is it important for the environment?

Renewable energy is important for the environment because it helps **protect nature and reduce pollution**.

Reasons

1. **Reduces air pollution** 🌿
   Renewable sources like solar and wind do not burn fuels, so they produce very little harmful gases.
2. **Lowers greenhouse gas emissions**
   They reduce gases like carbon dioxide that cause global warming and climate change.
3. **Conserves natural resources**
   Renewable energy uses sources like sunlight, wind, and water which **do not run out**, unlike coal and oil.
4. **Protects ecosystems**
   Using clean energy reduces mining and drilling, which can damage land, forests, and wildlife habitats.
5. **Supports a sustainable future**
   It ensures that future generations will also have energy without harming the planet.

✅ **In short:** Renewable energy helps keep the air clean, reduces climate change, and protects natural resources.

Now describe two examples of renewable energy sources.

Two common examples of renewable energy sources are:

1. Solar Energy ☀️

Solar energy is energy obtained from the **Sun**. Solar panels capture sunlight and convert it into electricity or heat. It is widely used in homes, street lights, and water heaters. Since sunlight is available every day, solar energy is a **clean and unlimited source of energy**.

2. Wind Energy 🌬️

Wind energy is produced using **wind turbines**. When the wind blows, it turns the blades of the turbine, which generates electricity. Wind energy is commonly used in wind farms to produce large amounts of power without causing pollution.

Both of these sources are **environment-friendly and renewable** because they do not run out and produce very little pollution.

4.Missing Word Prompting

Prompt 1 : "The brain of the computer is called the ____."

The brain of the computer is called the **CPU (Central Processing Unit)**.

So the completed sentence is:

"The brain of the computer is called the CPU."

Prompt 2 : "The process of converting data into a secret code to prevent unauthorized access is called ____"

The process of converting data into a secret code to prevent unauthorized access is called **Encryption**.